Hertenstein ecological health

A first master plan for measuring and improving the land at Stella Matutina

First draft · 13 October 2026 · internal working document

How to read this

This draft pulls together what public Swiss records say about the site, makes a first reading of what they mean, and sets out what to measure, how, and which changes to bring to landscape architects and permaculture designers. It ends with a readings system that visitors can use to build the record up over time.

Every claim carries one of three marks:

- **Checked**: read from a named public source during this pass (sources at the end).
- ***Inferred***: my reading of the checked facts, or general knowledge about this kind of landscape. Treat as a hypothesis.
- **Unknown**: not in the records I could reach. A site visit, a survey or a document from the owner is needed.

Two things frame all of it. The site changes hands through a sale that waits on official clearances, and the land is held and run by the Hertenstein side, not by Stella. Anything here that touches the ground is a proposal to plan with them, with the municipality of Weggis and with the canton. And the records are thin: no one has published a survey of this property, so most of what follows is a starting position for the baseline year, not a verdict.

Summary

The site. A former convent and education house at Zinnenstrasse 7 on the Hertenstein peninsula, on the north shore of Lake Lucerne, about 458 m above sea level and 24 m above the lake. It sits inside a federally listed landscape (BLN 1606, Lake Lucerne with Kernwald, Bürgenstock and Rigi), in a stretch the federal sheet describes as small-scale shore landscape with ice-polished rock hills, drumlins, beech woods, rich meadows and mostly walled shoreline. The Sisters of Baldegg held it from 1916. Their own description of the place mentions more than a hundred oaks, grazing animals, a garden being turned into a permaculture garden, and black kites that come back each year on 19 March.

What looks strong. The oaks. A park of mature oaks is the most valuable habitat a site like this can have, and it takes two centuries to replace. A white helleborine orchid was recorded within a few metres of the house in 2019. The shallow bays around the peninsula hold dense underwater meadows of stoneworts and pondweeds that the canton names as fish nurseries and bird resting places. The climate is mild for its latitude. Some buildings already carry green roofs.

What looks weak or at risk. The open land shows up in the federal habitat map as fertile hay meadow (Fromental meadow), which is usually a sign of years of fertilising and frequent cutting, and low plant diversity. The shoreline is mostly wall and riprap. Several invasive plants are recorded on or near the site (cherry laurel, Armenian blackberry, narrow-leaved ragwort, annual fleabane, a patch of black locust wood), as is the box-tree moth. In the lake, the quagga mussel arrived in 2024, the surface reached 26.3 °C at the end of June 2026, and by late July the lake level was a few centimetres above the point where outflow has to be held back. There are no public records at all of amphibians, mammals or bats in the surrounding square kilometres, which almost certainly means nobody has looked rather than that nothing is there.

First moves, in order:

1. Map and survey before changing anything. A baseline year, run by professionals for the hard parts (plants, birds, bats, trees, soil, shore) and by visitors for the repeatable parts.
2. Protect the oaks now: no digging, parking, trenching or building in their root zones, deadwood left standing where it is safe, and a tree survey to name the veterans.
3. Take the best-placed meadows off fertiliser and change the cutting regime, with clippings removed. This is the cheapest large gain available.
4. Put the edible landscape on ground that is already disturbed and close to the kitchens, not on the meadows or under the oaks.
5. Start removing the worst invasive plants, carefully, beginning with narrow-leaved ragwort and cherry laurel.
6. Open the shoreline conversation with Weggis and the canton. The owners' June 2026 statement already welcomes renaturation, and the municipality's own plan names a site on the peninsula, Postunentäli, as one of the best shore renaturation sites in the canton. Whether that ground falls within the property needs checking.

The readings system. Ten fixed stations around the grounds, each with a post and a QR code. A visitor scans it on a phone, picks a reading (a five-minute bird count, a ten-minute pollinator count, lake temperature, worms in a spadeful of soil, the night sky, the first kite of the year, an invasive plant), and sends it. Each reading becomes a row in a Sheet in the stella.earth Drive, with photos in a Drive folder. The protocol and a working form are in the same folder as this document and are ready to deploy.

1. The site from the records

1.1 Where it is

Address	Zinnenstrasse 7, 6353 Weggis (Hertenstein), canton of Lucerne	Checked
Coordinates (main building)	LV95 2 673 037 / 1 209 337; 47.0310 N, 8.3995 E	Checked
Ground height at the building	457.7 m	Checked

Lake level	about 434 m, so the house sits about 24 m above the water	Inferred from the usual mean level
Land register	Weggis parcel 286, EGRID CH382650357255, about 1.7 ha by its outline	Checked (area computed from the published outline)
Total holding	Other material speaks of 60 acres (about 24 ha). That would mean many more parcels than 286	Unknown
Owner and operator	Sale to a not-for-profit company still being formed, with a not-for-profit operating foundation; preliminary contract signed June 2026, sale completes after official clearances	Checked (Kloster Baldegg press release, 24 June 2026)

The list of parcels in the sale is the first document to ask for. Most decisions below depend on which meadows, woods and shore are actually in the holding.

1.2 Landform and geology

- The peninsula is a set of rounded rock hills (Tanzenberg, Grütschelen) shaped by the last glaciation, with drumlins and a "rib" landscape where harder layers of Molasse rock stand out. The federal BLN sheet names this landform as a protection goal in its own right. **Checked.**
- The geological map shows artificial fill under the buildings and Quaternary loose deposits around them. **Checked.** Away from the buildings, soils are most likely brown earths on glacial till over Molasse sandstone and conglomerate, shallow on the rock humps and deeper in the hollows. **Inferred.** No soil map or soil analysis for the property was reachable. **Unknown.**
- Wet hollows exist on the peninsula: the Hertenstein pond (a cantonally listed wetland) formed when a bay silted up, and the Postunentäli valley floor is described as moist with a culverted stream. **Checked.** Whether there are wet spots or drains on the property itself is **Unknown**.

1.3 Climate

- The Weggis and Vitznau shore has a mild climate influenced by the föhn wind (the federal BLN sheet). **Checked.**
- Lucerne station, 1991 to 2020: mean temperature about 10 °C, winter nights just below freezing, July days around 25 °C; rain about 1,200 to 1,300 mm a year, with the most in May to August. **Checked** from rounded figures republished from MeteoSwiss; confirm against MeteoSwiss's own normals table before relying on a number.
- The direction of change is hotter, drier summers, heavier single rain events and fewer frost days. 2026 already shows it on the lake (next section). **Inferred** from Swiss climate scenarios; the current national scenarios should be read for the numbers.

1.4 The lake in front of the site

- The canton's shore protection plan for Weggis lists two shallow-water areas at Hertenstein: the Hertenstein bay and the bay at the shipyard. Both have artificial shorelines, but underwater they hold stoneworts, Canadian waterweed and several pondweeds (five species in the shipyard bay, including Fries' pondweed). Both are fish spawning and nursery grounds (pike and carp family) and bird resting places. The conflicts named are boats in shallow water, the shipyard and illegal shore structures. **Checked.**
- Lake Lucerne's phosphorus levels are good to very good and its water quality is rated excellent. The surface reached 26.3 °C at the end of June 2026, close to the 2003 record. The quagga mussel was first found in the lake in 2024. In late July 2026 inflows had fallen by more than half and the level was a few centimetres above the threshold for holding back outflow. **Checked** (SRF, 29 July 2026).
- Invasive waterweeds (Canadian and Nuttall's waterweed) were recorded near the peninsula in 2021, and a mayfly of clean gravel shores (Ephemera danica) in 2024. **Checked** (GBIF).
- The quagga mussel matters for the site directly: it encrusts intakes, ladders and boats, strips plankton out of the water, and boats moving between lakes spread it. The cantons around the lake now require boats to be cleaned. **Inferred** from the general literature, plus press reports of the cleaning duty.

1.5 Habitats on and around the site

The federal habitat map (Lebensraumkarte, BAFU) within 40 m of the main building shows: **Checked**

- fertile lowland hay meadow (Fromental meadow), mapped with high confidence;
- grassland of unspecified type;
- trampled lawn and ruderal ground;
- a patch of **black locust (Robinia) wood**;
- **green roofs** on some buildings;
- buildings, sealed ground and an asphalt road.

The hectare-grid land-use survey (Arealstatistik) around the house records horticulture on two grid points, plus meadow, farm pasture and building surroundings, unchanged between the 1979/85 and 1992/97 surveys. **Checked.** The horticulture points fit the Sisters' kitchen garden and nursery. **Inferred.**

From the Sisters' own pages: more than a hundred oaks in a park-like setting, grazing animals, and a garden being built up into a permaculture garden over the two years before closure. The June 2026 press release says a garden group already works on the site and plans to grow it into an edible landscape that partly feeds the house. **Checked.**

Around the site, the canton's plan describes the peninsula as a varied landscape with valuable small woods (including sweet chestnut), orchards, hedges and single trees, and lists keep-free zones at Postunentäli, Tanzenberg and the chapel by the Bürgerheim. The woods between farmland in this part of the BLN area are mostly woodruff beech wood, with other beech types on steep lakeside slopes. **Checked.**

Not known yet: the area of each habitat on the property, the age and condition of the oaks, whether any meadow is still species-rich, the state of hedges and orchards, the soil, any drains or springs, and what the grazing animals were and whether they are still there.

1.6 Species already on record

GBIF holds 1,103 records in a box of roughly 2.5 by 2.2 km around the peninsula and western Weggis (47.02 to 47.04 N, 8.39 to 8.43 E): about 450 flowering plants (dicots), 330 birds, 185 monocots (grasses, orchids and the like), 58 insects, and a few fungi, ferns and mosses. **Zero amphibians, zero mammals.** Most records are from citizen science since 2020; older ones come from the national plant databank. **Checked.**

What stands out:

Group	Records	Why it matters
Orchids	White helleborine, 2019, located within metres of the main building. Lady's slipper (2021), green-winged orchid (2023) and bee orchid (2024), with locations deliberately blurred by the recording platform. Summer lady's tresses, autumn lady's tresses, late spider orchid and creeping lady's tresses recorded around Weggis between 1960 and 1983 at coarse precision	The helleborine is a woodland orchid of calcareous soils, likely under the oaks or a shady edge. The others suggest the area once held lean meadows and wet ground that are now rare. Lady's slipper and summer lady's tresses are protected across Europe. Exact locations should stay unpublished
Birds	Coot, moorhen, cormorant, mute swan, black-headed gull, mallard on the water; red kite and buzzard overhead; jay, blackbird, house sparrow, white wagtail	Common species so far, but only a sample was read. The Sisters' kites are most likely black kites. The BLN sheet names cirl bunting and peregrine falcon for the Rigi slopes
Insects	Box-tree moth (2024, 2025), death's-head hawkmoth (2024), the mayfly Ephemera danica (2024), various bugs and moths	Box-tree moth will strip any box hedges. No butterflies, bees or dragonflies on record at all, which is a gap, not an absence
Invasive plants	Cherry laurel (2024 to 2026), Armenian blackberry (2024), narrow-leaved ragwort (2024), annual fleabane (2025)	See 2.3
Amphibians, reptiles, mammals, bats	None	The federal sheet names the smooth newt and the threatened dragonfly Boyeria irene for this part of the BLN area. A survey year will fill this

The cantonal and national species centres (Info Flora for plants, info fauna for animals, the Swiss Ornithological Institute for birds) hold more than GBIF does, including precise locations of sensitive species. Those data come on request, normally free for conservation planning. **Unknown** until requested.

1.7 Protections and public plans that apply

- **BLN 1606, subarea 5 "western lake bays".** The site point lies inside it. **Checked.** The relevant goals: keep the small-scale shore landscape in its substance and character; keep natural underwater habitats and near-natural shores; keep the rib landscape and its geology. BLN binds federal and cantonal authorities when they carry out federal tasks (some permits, subsidies), so it will shape what gets approved, especially on the shore. **Inferred** from how BLN works.
- **The canton's protection plan for the Lake Lucerne shore, Weggis sheets. Checked.** For Hertenstein it lists:
- shoreline landscape protection along Postunen, Zinnen and Hertenstein, in front of the Villa Senar and Zinnen ensembles;
- Zinnen as a protected ensemble of 19th-century houses and boathouses with notable trees;
- the Hertenstein pond (cantonal wetland inventory object F 1), at risk from silting and nutrient run-off, with a buffer of extensive use proposed on the farmland around it;
- Postunentäli as one of the best shore renaturation sites in the canton: open the culverted stream, restore a natural land-to-water edge, extensify the valley floor. The landowner was recorded as opposed;
- a public path round the whole peninsula as the municipality's top path priority, and the old paths from Postunen over Zinnen to Hertenstein to be reconnected, which needs rights of way. The extract I read is not dated. Whether it still reflects the current zoning revision is **Unknown**.
- **The owners' public position.** The June 2026 press release says the new owners welcome further shore renaturation, want to keep the peninsula's landscape together with neighbours and the Rachmaninoff Foundation, and plan an edible landscape. The Sisters name *Laudato si'* as a reference for the project. **Checked.**
- **Zoning, building-line, hazard and groundwater rules for parcel 286** sit in the canton's land-register restriction extract, which I could not open from here. **Unknown.** This is a ten-minute lookup on the canton's geoportal.

1.8 History of use

- The Sisters of Baldegg founded a girls' institute at Hertenstein in 1916 and closed the education house in 2024. **Checked** (Historical Lexicon of Switzerland; Kloster Baldegg). Some Stella copy says the Sisters held the site "for two centuries"; the order itself dates from 1830, and its time at Hertenstein from 1916. Worth correcting in public copy.
- The hotel at the tip of the peninsula (now the Campus Hotel, a separate property at Hertensteinstrasse 156) goes back to 1864, hosted open-air theatre on the Tanzenberg from 1909 to 1921, and was rebuilt as a campus hotel in 2013. It owns 25,000 m² of forest and heats and cools with lake water. **Checked.** It is a neighbour and a useful precedent for lake-water heat.
- Before 1916 the land was most likely farmland, orchard and woodland like the rest of the peninsula. Ludwig II of Bavaria planned a large park here that was never built. **Partly checked** (the park plan is in the 2024 sound-walk release); the rest is **Inferred**.
- The national historical maps (swisstopo's Zeitreise series, back to the Siegfried map of the 1880s and the 1940s aerial photos) will show when woods, orchards, paths and buildings appeared or went. I could not view map images from here. **Unknown**, and the cheapest single piece of history work to do next: an afternoon on map.geo.admin.ch stepping through the years at this point and saving screenshots.

1.9 What I could not check, and how to fill it

Gap	How to fill it
Which parcels are in the sale, and their areas	Ask the owner side for the parcel list and plan
Zoning, building lines, natural hazards, groundwater zones	Canton of Lucerne geoportal, land-register restriction extract for each parcel
Historical maps and aerial photos	swisstopo Zeitreise; aerial image archive (LUBIS)
Soils	Soil samples and profile pits in the baseline year; ask the canton if any farm soil data exist
Precise records of plants and animals	Data requests to Info Flora, info fauna, the Swiss Ornithological Institute and the cantonal nature office
Tree age and condition	Arborist survey of the oaks and other large trees
Shore and underwater state	Shore mapping plus a diving or wading survey of the near-shore plants, coordinated with the canton
What the Sisters did with the land	A conversation with the sisters or staff who ran the garden and the animals, and the garden group now on site

2. First analysis

2.1 What kind of place this is

The site is a lakeside cultural landscape: a patchwork of hay meadow, oak park, small woods, hedges, garden and buildings on glacier-polished ground, with a walled shore facing shallow water that is rich under the surface. **Inferred** from 1.2 to 1.6.

Its ecological value lies mostly in three things that cannot be bought quickly: old trees, undisturbed soil under them, and the link between land and lake. Its weaknesses are the ones most Swiss lowland land shares: meadows kept too fertile, a hard shore, invasive garden escapes, and almost no recorded knowledge of what lives there. **Inferred.**

2.2 Strengths to build on

1. **The oak park.** Old oaks carry more species than any other native tree in central Europe: beetles that live in decaying wood, hole-nesting birds, roosting bats, lichens and fungi. Veteran trees with hollows and dead limbs are the core of this, and they are often removed for tidiness or safety. **Inferred.** First job: count them and name the veterans.
2. **Orchids and the soils they need.** The helleborine near the house shows calcareous, undisturbed soil somewhere close. The historic orchids around Weggis show what lean meadows here once held. **Inferred.**
3. **Shallow water rich in plants**, recognised by the canton. The site can add a softer edge and less disturbance. **Checked and inferred.**
4. **The setting.** It lies inside a national landscape inventory, close to a nationally important dry meadow (the Chestenenweid at Weggis, with its restored chestnut grove), and close to the federal aquatic research station at Kastanienbaum across the lake and the national bird institute at Sempach. Expertise is unusually near. **Checked** (the locations) and **Inferred** (that they would engage).
5. **Existing green roofs and good roofs for solar.** Some roofs are already green. Roof surfaces around the main building total about 1,190 m² in the federal solar roof data, of which about 400 m² are rated good or very good, with a modelled yield of roughly 75 MWh a year on those alone. **Checked** (sonnendach data, sampled around the main building only).
6. **A garden group already at work** and a public commitment to an edible landscape. **Checked.**

2.3 Pressures

1. **Over-fertile meadows.** Fromental meadow mapped with high confidence usually means regular fertilising and two to four cuts a year. Plant diversity in such meadows is typically low, and with it insects and the birds that eat them. **Inferred.**
2. **Invasive plants. Checked** records, with my reading of each: - *Narrow-leaved ragwort* (Senecio inaequidens): on the federal list of prohibited invasive plants, toxic to grazing animals and spreading fast along roads and dry ground. Priority one. - *Cherry laurel* (Prunus laurocerasus): sale banned in Switzerland since September 2024; birds spread it into woods, where it shades out the ground layer. Very likely planted as hedging on the site. Priority two. - *Armenian blackberry* (Rubus armeniacus): smothers edges and young trees. - *Annual fleabane* (Erigeron annuus): invades meadows, especially extensified ones, so it needs watching once the meadows change. - *Black locust* (Robinia): a mapped patch near the house. It fixes nitrogen and enriches the soil, which works against lean meadows nearby. It suckers when cut, so it is managed by ring-barking over two or three years. - *Box-tree moth*: kills box hedges; relevant for any formal planting on the site and for the boxwood hedge at the neighbouring Villa Senar.
3. **A hard shore.** Walls and riprap cut the gradient from land to water that young fish, insects and shore plants need. **Checked** that the shore is mostly artificial; the length on the property is **Unknown**.
4. **Nutrients running downhill.** The canton names nutrient input from farmland as the main threat to the Hertenstein pond. Whatever is spread on the property's slopes ends in the lake or the pond. **Checked and inferred.**
5. **Heat and drought.** A 26 °C lake surface in June and a lake near its intervention level in July point to summers that stress shallow-rooted planting, young trees and anything needing irrigation. Water use for the garden in July, when the programme runs and visitor numbers peak, needs a plan. **Inferred.**
6. **The summer programme itself.** A hundred and thirty to a hundred and fifty people across July means trampling, light at night, noise, boats and swimming at the shore, and pressure on paths. July comes after the main bird breeding season but when meadows are in flower and bats are raising young. **Inferred.**
7. **Light at night.** New outdoor lighting is a common, avoidable harm to bats and insects. **Inferred.**

8. **Quagga mussel.** Mostly a lake-wide problem, but any boats, paddleboards or diving gear Stella brings or hosts are a route for spreading it and other invasive species. **Inferred.**

2.4 Tensions to name early

These are choices, not technical problems, and the people at the table will see them differently.

- **Food versus meadow.** An edible landscape that feeds the house wants fertile, worked ground. Species-rich meadow wants lean, undisturbed ground. Both are possible on one site if they are kept apart. If the garden expands onto the meadows, the biodiversity case weakens. My suggestion is to grow food on ground already disturbed near the buildings and keep the best meadows lean.
- **Public path versus quiet shore.** The municipality wants a path round the peninsula; the canton's plan also wants a reserve at the pond where the path would pass. Stella's land may hold part of the route. An open, well-placed path is also how neighbours come to care about the site.
- **Tidy versus alive.** Deadwood, long grass, ivy on old trees and nettle patches look untended to some guests. A few signs and a clear mowing pattern (mown paths through long meadow) usually settle this.
- **Who decides.** Stella runs programmes; the land belongs to and is operated by the Hertenstein side. A measurement plan can start under Stella's programme; any change to the ground needs the owner's agreement and, for the shore and protected objects, permits.

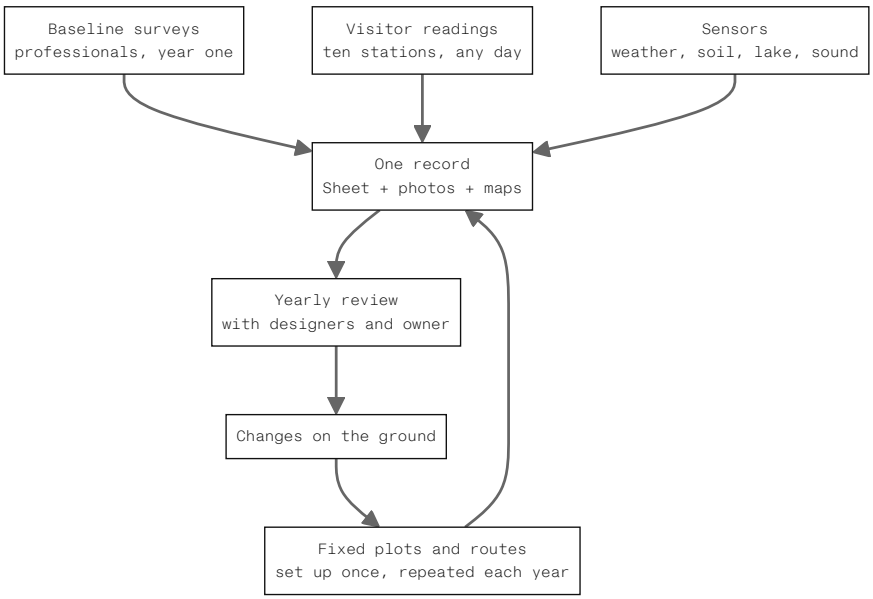
3. What "ecological health" means here

A short set of measures, each one something a landscape architect or permaculture designer can act on and a later year can be compared with. Targets are directions for now; numbers should come after the baseline year.

#	Measure	What it tells you	Direction
1	Meadow plant diversity: species in fixed 25 m² plots, and the federal list of meadow quality indicators	Whether meadows are getting leaner and richer	More species, more indicator species
2	Pollinators: timed counts on flowers along fixed routes	Whether the flowers support insects	More visits, more groups
3	Breeding birds: territory mapping in spring; point counts year-round	The whole system, from insects to cover	More species and territories, especially hole-nesters and hedge birds
4	Bats: passive acoustic recorders	Night-time health, effect of lighting	More species and activity
5	Old trees: count, size, condition, hollows, deadwood volume	The site's long-term capital	No loss of veterans; successors planted
6	Soil: organic carbon, pH, nutrients, structure, earthworms, infiltration	Whether management is building or depleting soil	More carbon and life, faster infiltration; nutrients falling in meadows
7	Shore and water: length of natural shore, near-shore plant cover, water temperature, clarity, pond nutrients	The land-to-lake link	More natural shore; stable plant beds
8	Invasive species: area and stem counts by species	Whether removal is working	Down
9	Darkness: sky brightness and counts of lit fittings	Light pollution	Darker
10	Site metabolism: water used, rain captured, food grown, compost made, energy generated	Whether the house runs on the site's own cycles	More closed loops

4. How to measure

Four layers, from expert to everyday.



4.1 Baseline year (professional)

Commission these in the first full season, ideally March to October 2027. Get two or three quotes for each; I have not priced them.

Survey	When	Who
Habitat map of the whole holding (to the federal habitat typology), with areas	May to June	Landscape ecologist
Vegetation plots: 10 to 15 permanent plots across meadows, oak park, woods and edges, marked with buried magnets and GPS	May to June, repeat in late summer	Botanist
Veteran tree survey: every oak and large tree, tagged, measured, condition and habitat features	Winter (leaves off)	Arborist with a habitat-tree method
Breeding birds: territory mapping, five or six visits	March to June	Ornithologist; the Swiss Ornithological Institute can advise on method
Bats: acoustic recorders at five or six points for a week in each of May, July and September; check buildings and old trees for roosts	May to September	Bat specialist (the cantonal bat protection network)
Amphibians and reptiles: pond and wet-spot checks, cover boards	March to June	Herpetologist
Insects: butterflies and grasshoppers along fixed routes; one deadwood beetle check of the oaks	May to September	Entomologist
Soil: a grid of samples for carbon, pH and nutrients; two or three profile pits	Spring	Soil lab plus a soil scientist
Shore and near-shore water: map the shore type along the property; survey underwater plants and mussels	July to August	Aquatic ecologist; the federal aquatic research station has a lab across the lake
Invasive species: map every patch	June and September	Can be done by the garden group with one expert day
Historical maps: step through the Zeitreise series and the aerial archive	Any time	Anyone, an afternoon

4.2 Fixed plots and routes

The baseline sets up things that are repeated in the same place, the same way, every year: the vegetation plots, a pollinator route, a bird point-count route, the bat recorder points, numbered trees, and shore photo points. This is what makes a later year comparable. Locations go in one shared map (a simple GIS layer or a Google My Maps map to start).

4.3 Visitor readings

Ten stations, each a post with a number and a QR code, set where the habitats differ: oak park, best meadow, ordinary meadow, garden, woodland edge, hedge, shore or landing, the pond or wettest ground if there is one, an orchard or the chestnut trees if present, and one open spot for the night sky. Visitors take short, structured readings that suit people without training, and photos do the rest. Full protocol in the companion document; working form in readings-form/.

Readings available:

1. Five-minute bird count
2. Ten-minute pollinator count on one flower patch (a simplified version of the UK's Flower-Insect Timed count)
3. Flowers in bloom within a one-metre square
4. Lake: water temperature, clarity, algae, mussels on stones
5. Soil: worms in a spadeful, how fast a litre of water sinks in
6. Night: how many stars, how many lights, any bats
7. First of the year: kites back, first swallow, oak leaves out, first frost
8. Weather snapshot
9. Invasive plant or pest seen
0. Anything else noticed

The readings are not a substitute for the surveys. They give the site a pulse between surveys, catch changes and seasonal events nobody would pay a specialist to watch, and give every visitor a reason to look closely at the place. Over five years, a few thousand readings at fixed stations become a real record.

4.4 Sensors

The site has been described as "sensor-ready". If so, a small set is worth more than many:

- one weather station (temperature, humidity, rain, wind, solar radiation), placed to national standards so it can be compared with MeteoSwiss;
- soil moisture and temperature probes in two meadow plots and one garden bed;
- a water temperature logger at the landing at about one metre depth;
- two or three passive acoustic recorders (birds by day, bats by night), rotated between stations.

The readings form has room for sensor values later; the simplest route is to keep sensor logs as separate files in the same Drive folder.

5. Priorities for landscape architects and permaculture designers

This is the brief to bring to the designers, in order. Each item says what it does, what the evidence is, and what it depends on.

5.1 Design principles

1. **Protect what is old before adding anything new.** Oaks, undisturbed soil, existing hedges and any lean meadow come first.
2. **Zone by intensity.** Intensive use near the buildings, extensive use further out. In permaculture terms, zones 1 and 2 (daily kitchen garden, herbs, nursery, compost, small fruit) sit next to the kitchens; zone 3 (orchard, food forest edge, any animals) on ground already worked; zones 4 and 5 (meadow, oak park, woods, shore) are managed for wildlife with light hands.
3. **Keep nutrients where they are wanted.** Compost and manure go to the garden only; meadows get none; nothing drains toward the lake or the pond.
4. **Work with water by holding it.** Capture roof water for the garden; slow run-off with planting rather than earthworks on the slopes, since the slopes face a protected landscape and a lake.
5. **Native and regional by default.** Plant regional provenance seed and stock; no new cherry laurel, box or other invasive ornamentals.
6. **Make the work visible.** Mown paths through long meadow, a few honest signs, and the readings stations show visitors that the long grass is a choice.

5.2 Ranked moves

#	Move	What it does	Depends on
1	**Oak protection zones.** Fence or mark the root zone of each veteran oak (roughly the crown spread plus a margin); no parking, trenching, storage or new paths there. Keep deadwood unless it is a hazard. Plant successor oaks from local acorns now	Secures the site's most valuable habitat for the next century	Tree survey; owner agreement
2	**Meadow restoration on the best-placed meadows.** Stop fertilising; cut twice a year (mid-June or later, and September), remove the clippings, leave 10 to 20 percent uncut each time as refuge. Where diversity stays low after two or three years, strip-seed with regional seed or spread green hay from a rich meadow nearby	Biggest gain for plants, insects and birds for the least money. Qualifies for federal biodiversity area payments if the land is farmed	Which meadows are in the holding; any farm lease; any grazing arrangement
3	**Edible landscape on disturbed ground.** Kitchen garden, herbs and nursery by the kitchens; a food forest edge and soft fruit between buildings and meadow; high-stem orchard trees of regional varieties where there were orchards before	Feeds the house, gives the programme a working garden, avoids sacrificing meadow	Garden group's existing beds; historical maps for old orchard sites
4	**Invasive removal programme.** Narrow-leaved ragwort pulled before seeding (gloves, bagged, not composted); cherry laurel cut and stumps removed or treated; Armenian blackberry cut back repeatedly; black locust ring-barked over two or three years; any box replaced	Stops the spread before meadows open up	Map from the baseline; a standing task for the garden group and summer volunteers
5	**Shore.** Join the municipality and canton on renaturation. On the property's own shore, look for one place where the wall can be stepped down to a gravel and reed edge; keep boat traffic out of the shallow plant beds; set a cleaning rule for any boats and boards Stella brings	The land-to-lake link the canton and the BLN both ask for	Shore length in the holding; permits; whether Postunentäli or the pond buffer touch the property
6	**A new small pond and wet ground**, away from the lake and fish, in a natural hollow if one exists, for newts, dragonflies and insects	Adds a habitat that is missing from the records	Soil and water survey; no impact on the protected landscape
7	**Hedges and small structures.** Native mixed hedges along boundaries and paths; stone piles and log piles on sunny edges; nest boxes for hole-nesters and swifts on buildings; bat boxes and bat access in roofs	Cheap cover and nesting, quickly	Bird and bat surveys first, so boxes go where they help
8	**Darkness.** A lighting rule: warm light, shielded downward, on timers or sensors, nothing lit toward trees, meadow or water	Protects bats and insects; costs little	Agreement with the operator
9	**Water and nutrients in the house.** Roof water to garden tanks; composting of kitchen waste; in time, a look at lake-water heat as the neighbouring hotel does	Lower water and energy draw, fewer nutrients leaving the site	Technical study; permits for lake-water use
10	**Solar on the good roofs**, alongside the existing green roofs (green roofs and solar panels can share a roof and cool each other)	Energy from the site	Heritage rules for the ensemble; roof condition

5.3 What to ask the designers for

- A site plan of the whole holding with habitat zones, protected root zones, intensity zones and the readings stations.
- A meadow and mowing plan, with a map of what is cut when.
- A planting palette of regional native species and regional fruit varieties, with no invasives.
- A water plan: roof catchment, storage, garden irrigation in a dry July.
- A shore concept to take to Weggis and the canton.
- A five-year phasing plan matched to the survey calendar, so each change can be measured.

6. Who to bring in

- **The owner and operator side**: parcel list, any farm leases, what the garden group is doing, who agrees to what.
- **Municipality of Weggis**: zoning revision, the peninsula path, the pond buffer, Postunentäli.
- **Canton of Lucerne, agriculture and forestry office (nature, hunting and fishing section)**: shore protection plan, renaturation funding, species data.

- **Info Flora, info fauna, the Swiss Ornithological Institute (Sempach, in the same canton)**: data requests and survey method advice.
- **The federal aquatic research institute's lab at Kastanienbaum**, across the lake: shore and quagga questions; a natural research partner for the labs.
- **Lucerne University of Applied Sciences and Arts**: already ran the "Inselträume" project with the Sisters on the peninsula's future.
- **The Rachmaninoff Foundation at Villa Senar** and other neighbours: shared paths, shared shore.
- **Local farmers**: who mows and grazes what, and on what terms.

7. Questions for Sylvan

1. Can we get the parcel list and plan for the sale, and any existing farm lease or grazing arrangement?
2. Who on the Hertenstein side should see this draft first, and is the garden group someone we can talk to directly?
3. Is a baseline survey year in 2027 something to budget for now, or should it be framed as a lab or research project with outside funding?
4. Should the readings system start this winter with a few stations (birds, weather, first-of-year, lake), or wait for spring and a full set of posts?
5. Do you want this turned into a short version for the landscape architects, or will you brief them yourself from this draft?

Sources

Checked during this pass, 13 October 2026.

- Federal geoportal (map.geo.admin.ch / api3.geo.admin.ch): address search, height, cadastral map (Weggis parcel 286), GeoCover geology, BLN inventory, BAFU habitat map (Lebensraumkarte), BFS land-use statistics, BFE solar roof data.
- BAFU, BLN object sheet 1606 "Vierwaldstättersee mit Kernwald, Bürgenstock und Rigi", 2017 revision. data.geo.admin.ch/ch.bafu.bundesinventare-bln/objectsheets/2017revision/nr1606.pdf
- Canton of Lucerne (lawa), Schutzkonzept Vierwaldstättersee, Koordinationsblätter Weggis. lawa.lu.ch/-/media/LAWA/Dokumente/njf/landschaften/lpvws/007_schutzkonzept_vierwaldstaettersee_koordinationsblaetter_weggis.pdf
- GBIF occurrence search, box 47.02 to 47.04 N, 8.39 to 8.43 E. api.gbif.org
- Kloster Baldegg: Stella Matutina pages (home, Ort) and press release of 24 June 2026. klosterbaldegg.ch/bildungshaus
- Historisches Lexikon der Schweiz, "Baldegg" (Kloster und Kongregation). hls-dhs-dss.ch/de/articles/012126
- Luzern Tourismus, press release on the Hertenstein sound walk, October 2024.
- Campus Hotel Hertenstein: history and sustainability pages. campus-hotel-hertenstein.ch
- SRF, "Sauber, aber bald trocken: Dem Vierwaldstättersee geht langsam das Wasser aus", 29 July 2026.
- MeteoNews, Lucerne climate normals 1991 to 2020 (from MeteoSwiss).